

Automatic Emotion Recognition for Groups: A Review

Emmeke A. Veltmeijer^{ID}, Charlotte Gerritsen, and Koen V. Hindriks^{ID}

Abstract—This article aims to summarize and describe research on the topic of automatic group emotion recognition. In recent years, the topic of emotion analysis of groups or crowds has gained interest, with studies performing emotion detection in different contexts, using different datasets and modalities (such as images, video, audio, social media messages), and taking different approaches. Articles are included after an innovative search method, including Dense Query Extraction and automatic cross-referencing. Discussed are the types of groups and emotion models considered in automatic emotion recognition research, common datasets for all modalities, general approaches taken, and reported performances. These performances are discussed, followed by an analysis of the application possibilities of the discussed methods. To ensure clear, replicable, and comparable studies, we suggest research should test on multiple, common datasets and report on multiple metrics, when possible. Implementation details and code should be made available where possible. An area of interest for future work is to build systems with more real-world application possibilities, coping with changing group sizes, different emotional subgroups, and changing emotions over time, while having a higher robustness and working with datasets with reduced biases.

Index Terms—Group emotion review, crowd mood survey, automatic affect recognition, sentiment analysis

1 INTRODUCTION

IN recent years, there has been an increased interest in methods for group emotion monitoring. This can be applied with several different goals in mind, such as surveillance, automated image or video annotation, and event detection [1], [2]. Current methods for group monitoring are often resource intensive and depend on human attention, easily overlooking important cues [3], [4]. Constructing a framework for monitoring emotions in a group is highly relevant, as emotions lie at the basis of behaviour of a group [4], [5]. Monitoring emotion, preferably in real-time, can be employed to predict behaviour and intervene when necessary [6].

Defining group emotion, a generally accepted view is that group emotion is established by two components, namely the bottom-up component and the top-down component [7]. Bottom-up components arise from the individuals and their emotions, while top-down components are context-dependent, thus depend on factors such as environment and group information [1], [8], [9], [10]. Group emotion can then be defined as the shared condition in a group that is a result of bringing together these two components [11]. In affective computing, the goal is to classify these group emotions as authentically as possible. The data modality and features extracted from the used data may pose a limit on the extent to which this is feasible. An example of this limitation

is the adoption of perceived emotion rather than experienced emotion as a ground truth.

Detecting group emotion is more complicated than detecting an individual emotion, as estimating the mood of a group is not necessarily similar to simply combining all individual emotions in that group, a topic on which extensive work has been done (see [12], [13], and [14] for some recent reviews on individual-based, rather than group-based, emotion recognition). In [9], a user survey was developed to investigate which factors influence group emotion perception. From this, it followed that both local and global features take part in the decision process of human annotators. One could thus conclude that techniques other than simply averaging individual emotions should be considered for group-level emotion detection [15], [16].

As far as we know we are the first to write a systematic review on this subject that discusses a broad range of topics that never before have been combined into one overview. The topics discussed are: group types and emotion models considered in the literature, datasets, general approaches taken and reported performances of the proposed methods. A survey paper by [17] discusses some core methods and datasets used for emotion estimation in crowds before 2017. Since then, the task of group emotion recognition has evolved into a broad research area, concerning many different crowds, emotions, techniques, and approaches. The goal of this review is to provide a clear overview of studies researching automatic emotion detection of groups as a whole. We consider all data modality types, such as visual data and audio data. ‘Group’ is a broad concept, as it can range from a small number of people to mass gatherings attended by thousands of people. We consider all types of groups, as long as the people forming the group are in each others physical presence. The main question this review aims to answer is the following:

- The authors are with the Department of Artificial Intelligence, Vrije Universiteit Amsterdam, 1081 Amsterdam, The Netherlands.
E-mail: {e.a.veltmeijer, cs.gerritsen, k.v.hindriks}@vu.nl.

Manuscript received 17 June 2020; revised 19 Feb. 2021; accepted 25 Feb. 2021.
Date of publication 12 Mar. 2021; date of current version 28 Feb. 2023.
(Corresponding author: Emmeke A. Veltmeijer.)
Recommended for acceptance by J. Hoey.
Digital Object Identifier no. 10.1109/TAFFC.2021.3065726

What is the current state of the field of automatic group emotion recognition?

To answer this question, the following subquestions are formulated:

- 1) What types of groups (dynamics, size, emotions) are considered?
- 2) What datasets are common in the field of group emotion research?
- 3) What bottom-up and top-down approaches exist for different data modalities?
- 4) Are performances comparable across studies?
- 5) To what extent are the discussed approaches applicable to real-world groups?

The remainder of this review is structured as follows: first, in Section 2, we describe the selection process of the articles that are discussed in this review. Section 3 describes the group types and emotion models used in different studies, answering the first subquestion. Section 4 discusses the datasets (subquestion 2) that are used in the literature. The third subquestion is discussed in Section 5, describing the general approaches taken in the articles. Section 6 describes the performances of all articles included in this review. Given this foundation, the discussion in Section 7 answers both subquestion 4 and subquestion 5, discussing performances as well as application possibilities. This review is concluded in Section 8. For a list of all abbreviations used in this study, we refer to Appendix A, which can be found on the Computer Society Digital Library at <http://doi.ieeecomputersociety.org/10.1109/TAFFC.2021.3065726>.

2 METHOD

In this section, the process of article selection for this review is described. The aim is to find articles describing group emotion prediction, however, there is not much prior knowledge of which keywords are most common in the field and will thus lead to the highest recall of relevant articles. We therefore start with many different queries, comprising as many possibly relevant terms as possible. The (based on title and abstract deemed) relevant articles that are found this way, are then used to find the keyword combinations that are used most often throughout these selected articles in an automated manner. We call this *Dense Query Extraction* (DQE). In this way, the resulting queries will be more objective rather than being based solely on personal keyword choices. The goal of DQE is two-fold:

- 1) Find queries that better capture the essence of the group emotion recognition studies, in an objective manner.
- 2) With these improved queries, find more relevant articles that were not yet found with the handcrafted queries.

These two goals can be summarized as follows: find as many relevant articles as possible, with as little queries as possible. This results in three stages: the initial article search (exploring), the DQE from the found articles, and the final filtering of articles found with these improved queries (exploiting). In this section the final article search will be described,

as following from the search queries found through the DQE. For details on the stages of the initial exploring article search and DQE leading to the final search queries, the reader is referred to Appendix B, available in the online supplemental material. A schematic overview of the process is shown in Fig. 1. Part A of this figure refers to the DQE and final article search with the retrieved queries, all selection done based on title and abstract of the articles. The shaded part of part A is discussed in Appendix B, available in the online supplemental material, the unshaded part in this section. Part B visualises the selection of articles based on the full text, followed by the automatic cross-referencing. All searches are done with Scopus,¹ as it is found to offer a wider range of journals than Web of Science [18], yet provides a number of articles feasible to manually select by having the option to search through title, keywords, and abstract. This as opposed to Google Scholar, where search results are either based on full texts or titles only, the latter being considered too limited (abstract search is possible, but only for the last year). A weakness of Scopus, as mentioned by [18], is that it is limited to articles published after 1995. However, since our research field is relatively new, first articles appearing from 2012 onward, this is not an issue.

2.1 Final Article Search

After obtaining the resulting 13 dense queries shown in Table 1 (see Appendix B, available in the online supplemental material, for substantiation of these queries), each of them is entered on Scopus. Table 1 shows the number of hits for each query, and the number of articles filtered from these. For all but four queries, most results come from the subject area ‘computer science’. When this is not the case, most results belong to the subject area ‘medicine’ or both ‘medicine’ and ‘psychology’ (bringing articles about illnesses or disorders, which are outside the scope of this review). These are then excluded to reduce the number of articles to be inspected. For the first query, this reduces the total number of articles from 4540 to 1286. Similarly, for the fourth, sixth, and seventh query this brings back the number of articles from 3042 to 1018, 89 to 17, and 201 to 119, respectively. Next, we start with a first-level selection, where articles are included or excluded based on title and abstract. The inclusion criteria are listed below:

- 1) Written in English.
- 2) Collective emotions of groups are recognized.
- 3) Individuals forming the group are in each others physical presence.
- 4) Emotion recognition is done in an automatic manner.
- 5) Performance of methods that are proposed is validated with real-life data.

First-level selection results in a total of 158 selected articles, as visualised in part A in Fig. 1. This search was done in the first two weeks of January 2020 and, to update the set of articles after the first round of peer review, in September 2020. This set includes articles from which it remains unclear if emotions are detected from a physical group or not. For these 158 articles, the full texts are downloaded if

1. <https://www.scopus.com>

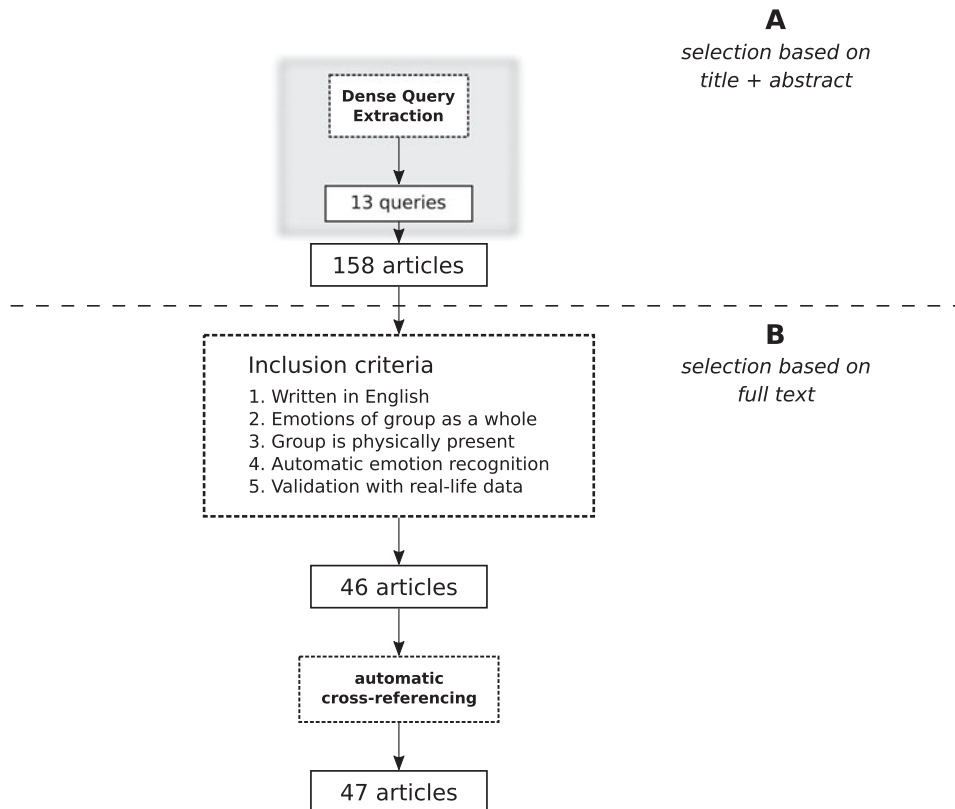


Fig. 1. Shown is a schematic overview of the article selection process, dividing it into selection based on the title and abstract of articles, article selection based on the full text, automatic cross-referencing based on the full text, and a second round of article searching. The shaded block in part A is discussed in Appendix B, available in the online supplemental material.

available (on January 13–14 2020 and in September 2020). The articles are then manually labelled using the Rayyan application for systematic reviews² [19]. For this second-level selection we consider the full texts. The same inclusion criteria as for the first-level selection apply. Reasons for exclusion, with the number of times applied, are:

- 1) Group emotion is not analysed (36 articles).
- 2) The group considered is not physically present or able to interact (44 articles).
- 3) Group emotion recognition is not done in an automatic manner (2 articles).
- 4) Methods are not validated with real-life data (e.g., simulation data only, acted emotions, or research proposals) (8 articles).
- 5) The article does not propose the methods it uses for group emotion recognition (rather they were proposed in earlier studies, e.g., reviews, or studies using existing techniques for a goal other than emotion detection) (7 articles).
- 6) Emotions are not measured from human factors, but rather influenced through external factors and predicted based on those (e.g., by changing scent or background music) (5 articles).
- 7) Methods for group emotion recognition are not described (9 articles).
- 8) Duplicate articles (different title to identical article) (1 article).

The first four exclusion criteria are similar to the (opposite of the) inclusion criteria. This means that for those articles, from the abstracts the inclusion criteria seems to be met, yet from the full text the opposite shows. These criteria result in exclusion of 106 articles, leaving a set of 46 included articles (from the 158 passing the first round). This is visualised in part B of Fig. 1. After this, automatic cross-referencing is performed by collecting all references used in all 46 included articles, and then carry out both first-level

TABLE 1
The Resulting 13 Queries Used in Our Literature Search, Total Number of Articles Per Query, and Number of Selected Articles

#	query	# total articles	# relevant articles
1	group emotion recognition	4540	54
2	crowd emotion analysis	188	21
3	crowd panic detection	32	15
4	groups emotion image	3042	37
5	disaster media sentiment analysis	119	26
6	group happiness intensity analysis	89	6
7	group emotion states detection	201	5
8	group-level scene affect recognition	6	4
9	group emotions sensor	156	9
10	crowd signal processing	450	5
11	audience emotion recognition	102	7
12	crowds sentiment media analysis	74	8
13	crowd emotions monitoring	19	4

The Number of Selected Articles Add up to More Than the Total Number of Articles is Discussed, as Some Appear in Multiple Queries.

2. <https://rayyan.qcri.org/>

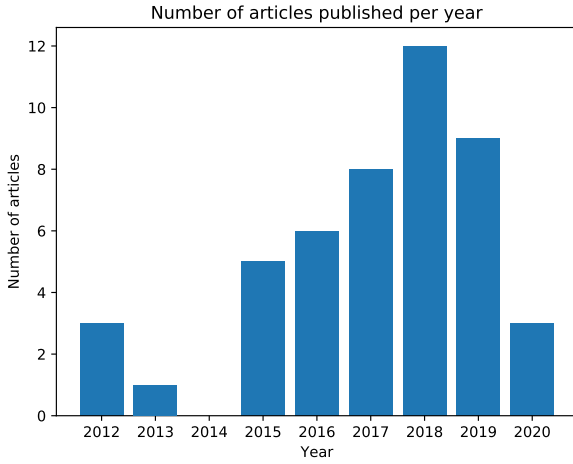


Fig. 2. The number of published articles in this review per year.

and second-level selection (as described above) on all articles that are cited by more than three articles of the set of 46 articles. This leads to the addition of one more article to the set of included papers, as visualised at the bottom of part B of Fig. 1. The final list of 47 included articles is listed in Table 2 in Appendix B, available in the online supplemental material. Fig. 2 shows the number of selected articles per publication year.

3 GROUP TYPES AND EMOTION MODELS

Having discussed the selection process of this review in the previous section, this section will focus on the first subquestion: *what types of groups (dynamics, size, emotions) are considered?* Studies working on group emotion detection have different definitions of both groups and emotions. A group, for example, can range from as small as four people together to as large as mass gatherings attended by thousands of people. Regarding emotions, different subdivisions of emotions are considered by the studies at hand, differing in the number of categories (from two to seven), number of emotions considered (from one to seven), and the emotion scale used.

3.1 Group Types

In this section, the group types that are investigated in the literature are discussed. They are ordered by increasing group size where possible. Group size, dynamics, and interaction possibilities are discussed. A general division can be found in Table 2.

The smallest group size number considered in an entire study is four. This is the case in [20], where groups of four people play the card game UNO. They are seated and thus a static (non-mobile) environment is created. There is a high amount of interaction between the group members, as they play against each other. A group size of around five individuals is found in [21] and [22], where five people are seated next to each other to watch videos. In the context of watching videos, there is not much room for explicit interaction between them. The group size is increased by one in [16]. In this study, an audience consisting of six individuals watches a presentation. It should be noted that there are more people in the room, but their data is not recorded (they do answer questionnaires), and that they are seated behind the six main participants. A work that investigates emotions of students in a classroom is [23].

TABLE 2
Group Types and Percentage of Studies in Which They are Employed

Group type	Description	Percentage
	Small, seated groups	10.64% (5 studies)
	Medium sized, mostly standing groups	76.60% (36 studies)
	Large, dynamic groups	12.77% (6 studies)

All groups considered so far are part of the first group type in Table 2: the small, seated groups.

The majority of papers discussed here work with images of social gatherings, including scenes such as parties, meetings, people posing together in different situations, people grieving, or demonstrations. Together these are used in as many as 36 studies, forming the second group type in Table 2 of medium sized, mostly standing groups (approximately up to 20 individuals),³ These images present rather static scenarios (people posing) interspersed with more ‘natural’ or dynamic (mobile) scenarios. One study [24] uses a combination of these social gathering images and images of people in explicitly non-controlled environments, and [25] contains videos that also show groups in non-controlled environments. As compared to the studies using social gathering images only (containing many posed images), this will not change the relationships between group members, but the introduction of non-controlled environments might result in more dynamic environments with more explicit interaction.

The group size is once more increased with the consideration of two studies gathering their data from surveillance videos. [26] consider outdoor videos, and [27] consider both outdoor and indoor video footage. These videos include people in a natural context, for example walking on the street or exercising. This makes the scenarios highly dynamic, in turn enabling explicit interaction. Four studies in this review investigate mass gatherings. In [4], the emotion around a stampede that happened after a boxing game is studied. [28] and [29] look at sporting events as well as other events that gather masses, such as riots. [30] investigate different city events, ranging from riots to celebrations. The number of people ranges from smaller gatherings (in specific locations) to mass gatherings (e.g., city-wide events). These scenarios can be seen as highly dynamic. These six studies employ the third group type in Table 2: large, dynamic groups.

3.2 Emotion Models

In this section, the different emotion models used in the literature are discussed. In the introduction it was discussed how the data (annotation) used in a study determines the emotion

3. All studies working with such datasets are shown in Table 8 Table 9 and Table 10 (some studies are present in multiple tables).

information extracted from the subjects under analysis, for example by annotating a ground truth that expresses the perceived rather than the experienced emotions. Differing in the exact emotion information gathered, the studies in this review have in common that they all base their emotion predictions on specific emotion models. The emotion categorisations in this review can be roughly be divided in derivatives of two different emotion models, discrete emotion labels [31] and arousal-valence emotion planes [32], and degrees of one emotion (happiness) only. They will be discussed in this order.

In 1971, Ekman and Friesen described six basic, universal emotions [31]. These are *happiness*, *sadness*, *anger*, *surprise*, *disgust*, and *fear*. Although critics have questioned this distinction, it has been widely adopted as a basic emotion classification in affective computing, among others because of the clearly distinguishable facial expressions. Seven studies use variations of these discrete emotion labels. [4] distinguish four of these, namely *anger*, *fear*, *happiness*, and *sadness*. They substantiate the omitting of *disgust* and *surprise* by referring to [33], in which it is stated that the facial expression of disgust is similar to that of anger, and that the same goes for surprise and fear. In [26] *surprise* and *disgust* are also omitted, and *excited* and *neutral* added instead. This follows the dataset they use (MED, see Section 4). [23] use Ekman's categorisation and add a *neutral* class to it. For their speech modality they leave out *disgust*, considering six emotions instead of the seven from the image analysis. In [20], the aim is to distinguish between *smile*, *surprise*, and *neutral*. The work of [28] considers the classes *joy*, *anger*, and *neutral*, and [29] consider *approval*, *disapproval* and *neutral*. In the work of [16], the goal is to detect the emotions *agreeability*, *interest*, and *fun* (and indirectly *boredom*, as being considered the opposite of fun).

Another emotion model is the Circumplex model, where arousal and valence are the perpendicular directions on a two-dimensional plane [32]. This arousal-valence (A-V) emotion plane was first proposed by Russell in 1980, with emotional states being represented by a continuous value along both axes. Arousal ranges from *low* to *high* with *neutral* in between, and valence ranges from *negative* to *positive*, also with *neutral* in between. The two-dimensional plane with eight concrete example values is shown in Fig. 3. This theory is applied in 30 studies, some studies incorporating one of the two dimensions instead of both. [34] predict both arousal and valence, each discretized to three values. Arousal can take on the values *high*, *medium*, and *low*, whereas valence can take on the values *positive*, *neutral*, and *negative*. [27] also consider both arousal and valence, and uses the full arousal-valence plane (with 10 steps on each axis) to plot a crowd mood curve. In [21] and [22] only arousal is considered, leaving the valence dimension out. This can be explained by recognizing their method, which considers bodily response of an audience to (audiovisual) stimuli. Hypothesizing that a moving audience is an aroused audience, they do not make any claims if this would be a positive or negative type of expressing. Other works consider only the valence dimension. This is done in 26 studies, that classify emotion as either negative, neutral, or positive. The high number of studies using this dimension can be attributed to a highly popular image dataset using it (GAFF, see Section 4). Two studies that use the

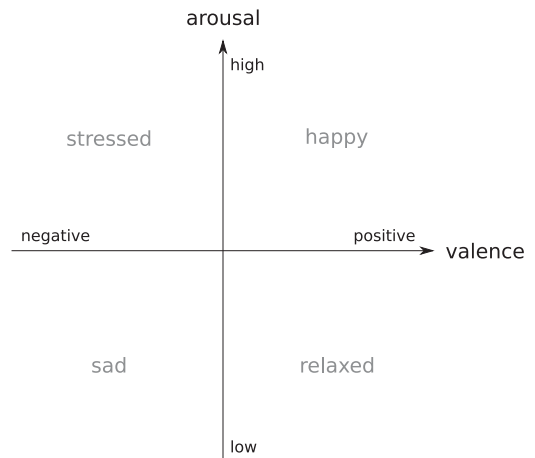


Fig. 3. The two-dimensional arousal-valence plane, with example states for each quarter of the plane.

valence dimension but do not work with this dataset, are [30] and [25]. For a list of all other studies using this categorisation, see Table 9.

Finally, there are 13 studies that focus on the prediction of one emotion type, namely happiness. These studies predict a happiness intensity score, ranging from 0 (neutral) to 5 (thrilled). For a complete overview we refer to Table 8.

Three studies incorporate combinations of the aforementioned categorisations in their work, or at least employ them during testing, by working with multiple datasets that each bring their own categories. [35] work with the positive/neutral/negative distinction (valence) as well as the happiness intensity score, both of which are also reported upon in [10]. [36] use these two classifications as well, and additionally report performance on an arousal-valence dataset (overlapping with the valence categorisation).

4 DATASETS

In this section popular datasets for group emotion recognition are discussed, thereby focusing on the second subquestion of this review: *what datasets are common in the field of group emotion research?* We will start with discussing datasets for the image modality. For group emotion detection from images, several datasets are used for training and testing. These can be roughly divided into group emotion datasets and (individual) facial emotion datasets. Next, video datasets and datasets of other modalities (such as audio) are described. For each modality, a table will summarize the datasets with their sizes and classes. A short description for each dataset is given. It is useful to separately mention the ImageNet dataset, as most of the Convolutional Neural Networks (CNNs) employed in this review (as well as in any type of computer vision research) are pre-trained on ImageNet [37]. This dataset includes more than 100,000 categories of everyday visual concepts and has proven to be an excellent dataset for pre-training.

4.1 Group Emotion Image Datasets

Group emotion image datasets will be discussed first, an overview of which can be found in Table 3. All datasets were published in the years 2012–2020, indicating the field

TABLE 3
Group Emotion Image Datasets Used in the Literature,
Including Year of Publishing, Number of Images,
and Emotion Classes Present

Dataset	# Images	Emotion classes
HAPPEI [38]	3134	Happiness
GAFF 2.0 [9], [39]	6471	Valence
GAFF 3.0 [9], [40]	17,172	Valence
MultiEmoVA [34]	250	Arousal and valence
EmotiC [41]	18,316	26 emotion labels plus arousal, valence and dominance
GroupEmoW [42]	15,894	Valence

is relatively new and active. For group emotion, two datasets are most popular: the HAPpy PEople Images, known as HAPPEI [38] (an example shown in Fig. 4a), and the Group Affect Database, known as GAFF [9] (an example shown in Fig. 4b). They are used in 13 and 24 of the studies in this review, respectively (some studies using both). The popularity of these can in part be attributed to their service as dataset for the EmotiW GREco subchallenge. HAPPEI was the official dataset for the challenge in 2016 [43], containing 3134 images, and GAFF in 2017 and 2018 [39], [40]. In 2017, GAFF (version 2.0) contained 6471 images, in 2018 increased to 17,172 (version 3.0). Both datasets show similarities and are proposed by the same authors, also organisers of the challenge itself. Images are scraped from Flickr for HAPPEI and from both Flickr and Google Images for GAFF, by searching with keywords related to groups and events. The HAPPEI dataset distinguishes six happiness intensities: *neutral*, *small smile*, *large smile*, *small laugh*, *large laugh*, and *thrilled*. These ground truths are given for the individuals in an image as well as for the entire image. The GAFF dataset contains ground truth categories *negative*, *neutral*, and *positive*.

Two studies use the MultiEmoVA dataset, one of them proposing it [34]. This dataset contains 250 real-life web images that are annotated on arousal (*low*, *medium*, *high*) and valence (*negative*, *neutral*, *positive*). It is collected from Flickr and Google Images with keywords indicating groups and events. Also used twice is the Emotion in Context Database (EmotiC) (once solely for pre-training), an example of which is shown in Fig. 4c. This dataset [41] contains people in non-controlled environments, distinguishing 26 different emotion labels. Additionally, valence, arousal, and dominance are labelled in a continuous manner on scales from 0 to 1.

TABLE 4
Facial Emotion and Facial Recognition Datasets Used
in the Literature, Including Number of Images and
Emotion Classes Present

Dataset	# Images	Emotion classes
FER-2013 [44]	36,887	Basic emotions
VGG-Face [45]	2,622,000	-
GENKI-4K [46], [47]	4000	Smiling/non-smiling
RAF-DB [48]	29,672	Basic and compound emotions
FER+ [49]	36,887	Basic emotions
SFEW [50]	700	Basic emotions

There are 18,316 images in total, with 23,788 individually annotated people.

Recently, a new group emotion image dataset was proposed by [42], called GroupEmoW. Images are scraped from Baidu, Bing, Flickr and Google, by searching with keywords related to groups and events. Each image is annotated as *negative*, *neutral* or *positive*.

4.2 Facial Emotion and Facial Recognition Datasets

Many of the studies discussed in this review, as can be read in Section 5, train networks to detect emotions from individuals. Although some group emotion datasets contain individual annotations as well, as was just discussed, there are many more facial expression recognition datasets that can be used for pre-training. An overview of facial emotion and facial recognition datasets can be found in Table 4. These datasets were published in the years 2009–2017. The two most common ones are the FER-2013 dataset [44], used ten times, and the VGG-Face dataset from [45], used seven times. The former contains 35,887 grayscale images of the following emotion labels: *happiness*, *sadness*, *anger*, *surprise*, *disgust*, *fear*, and *neutral*. Examples of each label are shown in Fig. 5. The VGG-Face dataset contains 2,622,000 images with 2622 individual identities, thus having 1000 images for each identity (public figures). This dataset is used for pre-training the corresponding face recognition model VGG-Face (a CNN used in many of the studies in this review). Other datasets used more than once for facial (emotion) recognition are GENKI-4 K (used five times, containing images of smiling and non-smiling ordinary people) [46], [47], RAF-DB (used three times, annotated with seven basic emotions and eleven compound, bimodal, emotions) [48], FER+ (used twice, an enhanced version of FER-2013) [49], and SFEW (used three times, distinguishing six basic emotions) [50].



(a)



(b)



(c)

Fig. 4. Example images from (a) HAPPEI [38], (b) GAFF 3.0 [9], [40], and (c) EmotiC [41].

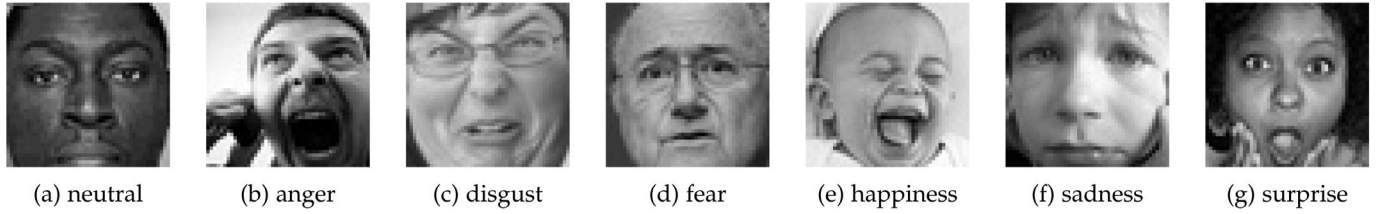


Fig. 5. Example images from FER-2013 [44].

4.3 Video Datasets

Of all studies incorporating the video modality, [21] and [22] are the only ones not using existing datasets or proposing one, but instead filming their own participants (for stimulus response graphs). An overview of datasets used by the other video studies in this review is given in Table 5. These datasets were published in the years 2009–2019. The UMN dataset is employed most often, twice in this review. This Unusual Crowd Activity Dataset of the University of Minnesota [51] contains 11 videos of escape events in indoor and outdoor scenes. Used once is the Performance Evaluation of Tracking and Surveillance 2009 dataset, known as PETS2009 [52]. This dataset contains crowd activity scenarios, some videos containing crowd panic behaviour. Example frames are shown in Fig. 6. In the work of [27], these videos are labelled with ground truths of arousal-valence values. Other video datasets that are used once are MED (Motion Emotion Dataset, containing five different behaviour classes [53]), Violent Flows (uncontrolled videos including violence [54]), and UCF (videos with normal and abnormal urban crowd scenes [55]). Lastly, the Video-Level Group Affect (VGAF) database is proposed in [25]. Videos are collected from Youtube and contain people in differing settings and group sizes. Each video clip is labelled as *negative*, *neutral*, or *positive*.

4.4 Other Data Modalities

Next to image and video data, studies discussed in this review also work with other data modalities. For textual analysis, an overview is given in Table 6. These datasets were all published in the years 2011–2015. Ngo *et al.* [4] employ the NRC hashtag emotion lexicon [56], which is labelled with *anger*, *fear*, *anticipation*, *trust*, *surprise*, *sadness*, *joy*, and *disgust*. In [30], three different datasets are employed. The first is the University of Michigan Sentiment Analysis competition dataset, containing social media posts that are labelled as either *negative*

or *positive* [57]. They second work with the Twitter sentiment corpus of Niek Sanders, containing random tweets with labels *negative*, *neutral*, and *positive* [58]. These two together are referred to as common-based datasets. Third, they create their own event-based dataset, containing tweets and Instagram posts from selected city events from a range of different activities and group sizes. Sharma and Mansotra [23] create their own dataset of tweets that are related to classrooms and express comments, feedback, or opinions.

In the work of [16], focusing on pressure sensors, they form their own dataset of center of force measurements. For the analysis of audio, both [28] and [29] create their own dataset as well. Both datasets contain sound clips from real events, including cheering (for *joy*), rioting (for *anger*) and background noise (for *neutral*). ‘Fake’ audio, such as from movies, is excluded to only include real emotions and not simulated ones. In [23], an unspecified emotional speech corpus is used for speech analysis, which includes group conversation and predicts six emotion labels. Lastly, [20] also construct their own dataset by recording groups of four people playing the boardgame UNO. Each participant has a microphone and camera directed at his or her face.

5 APPROACHES

In this section we will discuss the approaches studies take to automatically recognize group emotion, focusing on the third subquestion of this review: *what bottom-up and top-down approaches exist for different data modalities?* For this we will distinguish bottom-up approaches, top-down approaches and hybrid approaches.

Both bottom-up and top-down, sometimes referred to as local context and global context respectively, are commonly used in literature. In [7], Barsade and Gibson described them as the components of group emotion. A popular implementation of this distinction in affective computing is that described in [8]: bottom-up is considered to involve the emotion of the individual, and top-down takes into account context such as the background of an image. We adopt this distinction, with the adjustment that all features retrieved per individual, not just those regarding individual emotions, are considered bottom-up. An example of a feature changing category due to this adjustment is that of relative face size: in [8], this is incorporated in the global context (top-down) part of the proposed model, whereas in this review it is considered a bottom-up feature as it arises from the individual.

We will first consider research that has a bottom-up approach, followed by top-down approaches and methods that combine bottom-up and top-down approaches, referred to as hybrid approaches. When a study proposes several methods (within one approach type), we only include the

TABLE 5
Video Datasets Used in the Literature, Including
Emotion and Behaviour Classes Present

Dataset	# of videos	Emotion/behaviour classes
UMN [51]	11	Panic behaviour
PETS2009 [52]	126	Crowd activity including panic behaviour
MED [53]	31	Five behaviour classes
Violent Flows [54]	246	Violent behaviour
UCF [55]	20	Normal/abnormal urban crowd scenes
VGAF [25]	1004	Valence-labelled clips

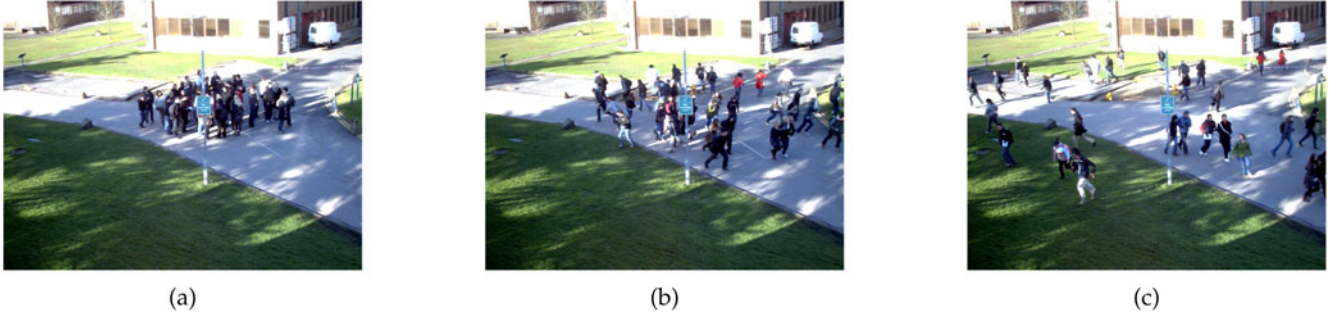


Fig. 6. Example frames from an escape event in PETS2009 [52].

best performing one, unless a comparison between the used methods is deemed relevant for the discussion. Additionally, methods will be discussed for fusion of different modalities, fusion of different analysis aspects (for example combining facial and scene analysis) and fusion of individual features and emotions to determine the common group emotion.

5.1 Bottom-Up Approaches

Bottom-up approaches work from an individual level to a global level, considering individual emotions or features, as indicated in Fig. 7. Data types that are used to provide individual information are images, audio, and social media messages, either separately or combined in a multi-modal approach. The step to the global level can be taken either on the decision level, after forming emotion estimates of individuals, or on the feature level, forming the group emotion estimate from individual features rather than individual emotions. Each modality type will be discussed here, starting with images and videos. The focus lies on the individual feature and emotion extraction, the fusion of individual information will be discussed in Section 5.4.

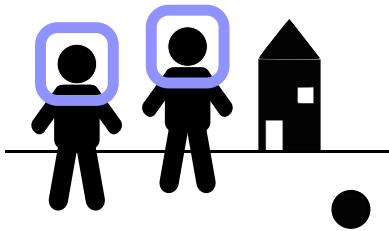


Fig. 7. Graphical representation of the bottom-up approach, defined in this review as using individual (emotion) features to reach a group emotion estimate.

TABLE 6
Text Datasets Used in the Literature, Including
Emotion Classes Present

Dataset	# of messages	Emotion classes
NRC hashtag emotion lexicon [56]	21,051	Basic emotions
University of Michigan Sentiment Analysis [57]	40,138	Valence
Twitters sentiment corpus of Niek Sanders [58]	5513	Valence

5.1.1 Images and Videos

Of the 47 studies discussed in this review, 12 work with bottom-up visual analysis only. Some studies include global information about the faces, such as their position or size, to combine the individual emotions. Since the group emotion is then still based on the individuals in the image, in this review, we consider them to take a bottom-up approach. Details on the exact descriptors and neural networks used in each study can be found in Table 1 in Appendix C, available in the online supplemental material.

Classic descriptors, i.e. non-neural approaches using for example Facial Action Units (FAU) ⁴, are used in 5 of the 13 studies discussed here. The resulting face descriptors lead to an individual emotion estimate through regression [38], [60] or Support Vector Machines (SVM) [61]. Huang *et al.* [62] propose Riesz-based Volume Local Binary Patterns as face descriptors (RVLBP), modelling group emotion with Continuous Conditional Random Fields while incorporating the size of faces and their relative distances. Descriptors from CNNs are used in two studies using image data only. In both, the individual face features are directly fed to a classifier to predict the group emotion, rather than predicting individual emotions first. This is still considered bottom-up, as the final prediction is based on the features of individual faces. These classifiers employed are either a combination of non-neural classifiers [63] or a combination of a Long Short-Term Memory (LSTM) and a dense layer [64]. In [65], CNNs are employed for emotion prediction for individual faces, fusing the individual predictions to form a group prediction. Combinations of classic descriptors and features from CNNs are used in two studies. In both studies, these individual-based descriptors are directly mapped to a group emotion, either by an SVM [36] or via Multi-view Common Space Learning, mapping all separate features to the common space of the output classification [66].

Finally, two studies do not work with distinct face descriptors, but incorporate other techniques to predict emotions from the input data. In [21] and [22], upper bodies of participants are recorded while they are watching videos. Movement synchrony is considered to express arousal, therefore the ratio of individuals responding to a stimulus to all individuals forming the group stands for the arousal measurement. In [67], individual faces are fed to multiple

4. Action Units (keypoints) of the Face Action Coding System (FACS), as proposed by [59] describing facial expressions by facial movements.

CNNs for individual emotion prediction. These predictions are then combined in heat map images (containing one heat map for each face), which are again fed to a CNN for the final group emotion prediction.

5.1.2 Social Media Messages

Two studies incorporate data from only social media in their work. The earliest of these is the work of [4], in which Twitter data is collected from people participating in mass gatherings, by filtering on geo-location and hashtags. Each tweet gets assigned a four-dimensional emotional status vector. The dimension in which the vector is largest corresponds to the emotion to which the tweet is mapped. From this, an emotional distribution of the crowd over time is constructed. Gong *et al.* [30] derive sentiments of crowds at city events from social media, proposing an event-based dataset. Emotion estimation is done both with lexicon-based methods and (non-neural) machine learning methods, the latter outperforming the former.

5.1.3 Multi-Modal Approaches

Two studies in this review work with multiple modalities, which in this review is considered as using different data modalities. In [20], emotions are recognized based on face and speech information. The goal of the speech information is to estimate group emotion during conversation. Power and pitch of the audio data are used as prosodic (element of speech) features. For the visual data, facial features are extracted and lead to individual emotion prediction through K-Nearest Neighbours (KNN). In [23], visual, audio, and social media data from a classroom are used for student emotion detection. From individual faces, descriptors from CNNs are used for emotion prediction. Mel Frequency Cepstral Coefficients (MFCC) is used for extracting audio features, after which the data is converted into tensors. An LSTM RNN is trained on an emotion corpus dataset. For social media data, they use messages from Twitter, and employ a lexicon-based method for emotion analysis. Features from this are then fed to a learning-based approach, an SVM classifier.

5.2 Top-Down Approaches

We will now discuss studies that use a top-down approach, analysing data in its entirety, as indicated in Fig. 8. Data types that are used to provide this group-level information are images, videos, audio, and pressure sensors. Each modality type will be discussed here, starting with images and videos.

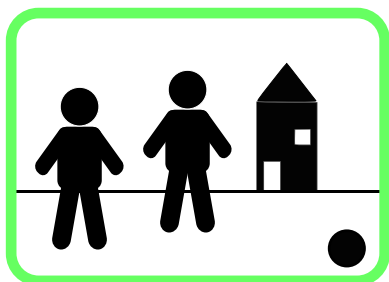


Fig. 8. Graphical representation of the top-down approach, defined in this review as employing scene-level analysis to reach a group emotion estimate.

5.2.1 Images and Videos

Of the 47 studies discussed in this review, four work with top-down visual analysis, three of which consider video data (one works with a multi-modal approach and is therefore discussed in Section 5.2.4). The only study using images is that of [43], where the baseline for the GRECO subchallenge of EmotiW 2016 is presented. A CENTRIST descriptor is employed to extract features from the images, which are then used for classification by support vector regression.

Zhang *et al.* [27] work with video data for group emotion recognition. Emotion motion features are extracted from coherent motion patterns, for both arousal and valence separately, and mapped to the arousal-valence plane. This results in a crowd mood curve to closely monitor a crowd. Video data is also used in [26]. Proposed by them is a 3D CNN to learn high-level spatio-temporal features for emotion detection. The third dimension, forming the temporal dimension of the CNN, implicitly captures motion. Details on the exact descriptors and neural networks used in each study can be found in Table 1 in Appendix C, available in the online supplemental material.

5.2.2 Audio

Two studies in this review perform group-level sound analysis only. Franzoni *et al.* [28] aim to detect crowd emotions from speech and sound in mass events, extending their work in [29]. To this end, sound spectrograms are used as input to a CNN (pre-trained on an image recognition task), where the spectrogram images are generated from snippets of sound clips with voice-based and clatter-based sounds (see Table 1 in Appendix C, available in the online supplemental material, for details about the network). The authors hypothesize that individual sounds are less important than the overall sound coming from a crowd, when the goal is to detect the emotion of that crowd in its entirety.

5.2.3 Pressure Sensors

One study in this review makes use of pressure sensors to analyse group emotions. In [16], an audience watching a presentation is seated in sensing chairs that measure the center of force for each seated individual. The relation between (synchronous) body sways and emotions of the audience is analysed, finding that boredom (and thus its antonym fun) can indeed be estimated from synchrony of body sways.

5.2.4 Multi-Modal Approaches

One study in this review works with both the visual and audio aspect of video, thereby taking a multi-modal approach in a top-down fashion. In [25], the image frames and audio fragments from video clips are processed separately, after which the features are combined to form a single emotion prediction. A CNN is used to extract visual features from each video frame, consecutively fed to an LSTM to get one visual feature per video clip. From the audio fragment, features from the OpenSMILE [68] toolkit are extracted and fed to fully-connected layers to gain the audio feature of each video clip.

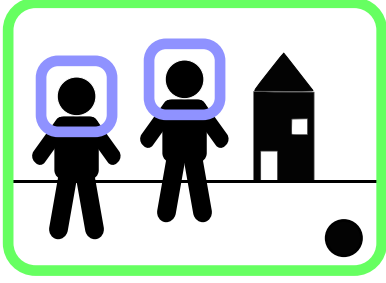


Fig. 9. Graphical representation of the hybrid approach, defined in this review as combining both bottom-up and top-down approaches.

5.3 Hybrid Approaches

This section will describe studies with a hybrid approach: joining bottom-up and top-down approaches, as indicated in Fig. 9. In this review, a hybrid approach is taken with images only, in 24 of the 47 reviewed studies. Combining approaches can be done focusing on different aspects of an image: for example by employing information from faces and entire scenes, or also including body information. We will discuss these studies on the basis of the aspects that they focus on. Details on the exact descriptors and neural networks used in each study can be found in Table 1 in Appendix C, available in the online supplemental material.

Most studies combine face-level analysis with scene-level analysis. Of these studies, [9] are the only ones working solely with classic descriptors, for both faces and scenes. For newer studies, it appears to be more common to either combine a classic approach with a neural approach, or to take a neural approach only. The former is done in the research of [69], [70], and [71]. In the first two of these, (best) performances come from CNNs for facial analysis and classic descriptors for scene analysis. Rassadin *et al.* [71] incorporate CNN-based and classic face feature extractions, combined with a CNN for the scene-level analysis. Surace *et al.* [72] use scene descriptors as nodes in a Bayesian Network, in addition to a CNN for faces (also functioning as input to the Bayesian Network). A study by [73] uses CNNs for both faces and scenes, but adds two classic descriptors (next to a CNN) for the entire image when no faces are detected. Other studies that work with CNNs on both the aspects of faces and scenes are [1], [15], [74], [75], [76], and [77].

Khan *et al.* [78] distribute these two aspects (faces and scenes) over four different streams. They train their networks on faces, images with attentional heat maps for each face, images with blurred faces, and the entire images without additions. In [10], separate neural networks are trained for facial features and for facial expressions, next to a CNN scene analysis. Not all images are fed to the scene-level CNN in the work of [79]. First, a face-level CNN and SVM distinguish between positive and non-positive (neutral and negative combined) images. The authors state that, based on an investigation of the dataset, positive emotions are easiest to distinguish. Only the non-positive images are then fed to a scene-level CNN, to categorize them more carefully into neutral, negative and (harder to distinguish) positive.

Nine of the studies taking a hybrid approach incorporate other aspects next to, or instead of, faces and scenes. Visualisations of these approaches can be found in Table 7. Upper bodies are used next to faces in the research of [35] and [34], the

TABLE 7
Hybrid Approaches Considering Aspects Next to, or Instead of, Faces and Scenes, and the Studies That Employ Them

Aspects	Description	Studies
	Faces, scenes, and upper bodies	[34], [35]
	Faces, scenes, and bodies/skeletons	[80], [81], [82]
	Faces, scenes, skeletons, and visual attentions/objects	[2], [42]
	Faces and objects	[83]
	Faces, scenes, and places	[24]

former also considering a classic scene descriptor and the latter also considering a context descriptor. Both use classic descriptors for faces as well as upper bodies. In [80], CNNs are used for analysis of faces, scenes, and bodies, and [81] and adds a skeleton analysis to the face and scene analysis, all done with CNNs. Faces, scenes and skeletons are also analysed with CNNs in [82], where on the face-level the CNN output is fed to an LSTM, and where on the scene-level an attention mask is placed over the image. Attention is also applied in [2], where next to faces, scenes, and skeletons, visual attentions are included (salient regions important for emotion detection, found by neural attention mechanisms) by feeding 16 salient regions to a CNN and LSTM. In the work of [42] the same aspects are considered, replacing visual attentions with objects. Each aspect is first fed to a CNN, from which the features are used as nodes in a complete graph. This graph is then updated for a certain amount of time steps, enabling the features from different aspects to interact. Next to analyzing faces both with a classic descriptor and a CNN, the work of [83] also proposes the analysis of objects for group emotion detection (through CNNs). Lastly, a comparative study is done by [24], where CNNs are employed to extract group emotion from faces, scenes, and places. These outcomes are not fused, but compared, to investigate the performance of face-based methods (faces, scenes) to the performance of non-face-based methods (places, for scene recognition).

5.4 Fusion Methods

To recognize group emotion, many studies use some form of fusion. This can be fusion of different modalities, fusion

of different analysis aspects (for example combining facial and scene analysis), or fusion of individual emotions in a bottom-up approach.

5.4.1 Fusion of Different Modalities

Three studies in this review take a multi-modal approach. The earliest research is that of [20], where a Bayesian Network is constructed for group emotion estimation. Individual emotions at each time step are a combination of face and speech information, each individual being represented by a node in the network. In [23], decision-level fusion is employed based on the probabilities, accuracy, and precision rates of the individual modalities (face, speech, text). [25] use feature-level fusion by concatenating their visual and audio features. The concatenated feature leads to a final emotion prediction through a softmax activation function.

5.4.2 Fusion of Different Analysis Aspects

When analysing different aspects of an image, these aspects have to be fused to form a joint group emotion prediction. Exceptions to this are [79] and [24]. In [79], images are first analysed based on their individual faces, and based on that prediction consecutively fed to a scene-level network. Nagarajan and Oruganti [24] aim to compare different modalities rather than fuse them.

The most popular approach towards fusion of different aspects, is to fuse individual predictions in a weighted manner. This is done in [81], [2], [78], [71], and [77]. A combination of individual predictions is also used in [34], [80], and [42], the latter employing majority voting. One or more fully-connected layers for fusion of individual aspects is employed in [1], [69], and [10]. An LSTM for feature fusion is adopted in [15] and [70], whereas [83], [82] and [35] use an SVM, the latter with Revisited Localized Multiple Kernel Learning (MKL). MKL is also proposed for fusion (and prediction) in [9]. Other fusion methods used are concatenation followed by KNN [75], a fusion network [73], a Bayesian Network [72], or a combination of weighted feature fusion (fusing local information with global information) and a Random Forest (fusing the fused features from different CNNs) [76]. Lastly, in [74], features are concatenated, but the step leading towards a common emotion prediction is not described.

5.4.3 Fusion of Individual Features and Emotions

When individual emotions are predicted or individual features are extracted, in bottom-up and hybrid approaches, these have to be fused to form a single group emotion. A rough distinction can be made between feature-level fusion and decision-level fusion. With feature-level fusion different features (in this case of different individuals) are fused before classification, whereas with decision-level fusion each feature (in this case individual) gets its own classification, and these outcomes are then combined.

Decision-Level Fusion. Fusing at decision-level can be undertaken in several ways. Some approaches take the average over all individuals, such as [62], where the individual weighted happiness intensities (the label with the highest probability for each face) are averaged. An average is also taken in [2], [72], and [77], where the emotion class with the resulting

highest probability is chosen as the group emotion. In [66], each face gets a confidence score for each emotion. To get a final happiness prediction, the summation of each possible intensity times its corresponding confidence score is rounded to the nearest intensity. Ghosh *et al.* [10] perform group-level pooling over the faces, and majority voting is employed by Guo *et al.* [42]. Other studies weigh faces, often according to their presumed importance for the image as a whole. A weighting scheme is used in [38], [81], [78], [61] and [65].

Other studies take a machine learning fusion approach, such as the work of [60]. They experiment with different fusion techniques, getting the best performance from feeding the mean and distribution of individual emotions in the image to an MLP to give the final group emotion. In [67], heat maps are generated for each face, indicating facial size and emotion intensity, with multiple CNNs. These are averaged per face, and then stacked per image (each heat map being located at the original face location). The stacked images, with the heat maps of all individuals, are fed to a CNN for the final group emotion prediction. Cerekovic [69] feeds the individual emotion predictions, together with information about their location and size, to an LSTM for the group emotion classification. As an LSTM takes input in a sequential manner, this simultaneously solves the problem of a varying number of individuals per image. A Bayesian approach is taken in [20], where individual emotions (resulting from face and speech features combined) influence group emotion through a Bayesian Network. Weighting with a neural touch is done in the final three studies to use decision-level fusion: in [79], [74], and [80], each individual emotion gets assigned a weight that is given by a neural network.

In the studies using social media messages, features are extracted at an individual level, and therefore also have to be merged to get to a group emotion. In the work of [4], the total score for each emotion is the proportion of tweets being assigned to that emotion. Gong *et al.* [30] does not fuse individual emotions, but instead calculates a group emotion error estimated from the individual emotions.

Feature-Level Fusion. Next, feature-level fusion mechanisms are discussed. In these studies, features are in some way combined before being fed to a classifier. Liu *et al.* [75] use a simple average of face features, and [63] also average individual embeddings to get one image feature. In [21] and [22], group arousal is calculated from the ratio of responding individuals to all individuals. This is similar to taking the average of a binary response variable (an individual responding or not, given a certain threshold). Rassadin *et al.* [71] construct a feature vector by taking the median of all individual face features, while [76] take a weighted average with the weights depending on the face size. This is an approach that is also common on decision-level, as described previously. A visual vocabulary is used in [9]. Representing each face in the image by a word from the dictionary, with a set dictionary size, resolves the issue of the variable number of faces per image. Huang *et al.* [35] propose INformation Aggregation (INFA), to stack the features of individuals. Balaji and Oruganti [83] experiment with both VLAD [84] encoding and Fisher Vector encoding [85]. In [36] an SVM is proposed for classification, with Combined Global Alignment Kernels that work irrespective of the number of faces. A popular approach, one we also

TABLE 8
Results and Approach Details of Articles Working With the HAPPEI Dataset

#	Article	Metric	Perf. on test	Perf. on val	Perf. on cross-val	Approach	Classic/CNN features
1	[70]	RMSE	0.822	0.486	-	Hybrid	Both
2	[60]	RMSE	0.83,163	0.517	-	Bottom-up	Classic
3	[64]	RMSE	0.836	-	0.4942	Bottom-up	CNN
4	[66]	RMSE	0.9525	0.3666	-	Bottom-up	Both
5	[43]	RMSE	1.3	0.78	-	Top-down	Classic
1	[10]	RMSE	-	0.26	-	Hybrid	CNN
2	[69]	RMSE	-	0.55	-	Hybrid	Both
3	[65]	RMSE	-	0.58	-	Bottom-up	CNN
1	[35]	RMSE	-	-	0.4516	Hybrid	Classic
2	[61]	RMSE	-	-	0.75	Bottom-up	Classic
1	[64]	MAE	-	-	0.228	Bottom-up	CNN
2	[35]	MAE	-	-	0.4821	Hybrid	Classic
3	[36]	MAE	-	-	0.492	Bottom-up	Both
4	[62]	MAE	-	-	0.5292	Bottom-up	Classic
1	[38]	MAE	0.379*	-	-	Bottom-up	Classic
1	[64]	Acc	77.94%	-	-	Bottom-up	CNN

Shown are Performances on the Test Set, Validation Set, or After Cross-Validation.

* : Unclear on Which Images Tested.

saw at the decision-level, is the employment of LSTMs for aggregating individual faces. In [15] they use an LSTM for both face and scene features. The five biggest faces are used, performing zero-padding if there are less than five faces detected. Other studies, making use of the sequential nature of LSTMs to take a variable number of faces, are [70], [82], [64], and [73].

A multi-modal approach that does not discuss individual emotion fusion is [23]. Studies that do not include information about individual emotion fusion (but are included because they also work with other aspects, such as global image features), are [1], [34], and [24].

6 PERFORMANCE

Having discussed the methods used by different studies in the previous section, this section lists performances of the approaches where possible. To provide a clear overview, studies are sorted by the datasets used by them. This will lay the foundation for the discussion of performances in Section 7.

The results of the best performing models of each study in this review are shown in Tables 8, 9, 10, and 11. Each table includes articles using a particular dataset, the last table (Table 11) containing datasets that are used once. If an article reports performance on multiple datasets or multiple metrics, then it is present in all respective tables. All tables include the reported metrics and performances, along with approaches (bottom-up, top-down, or hybrid). Performances within a table that are directly comparable are put together in a block (different blocks separated by horizontal lines). Each table will be explained separately.

6.1 HAPPEI Dataset

In Table 8, results reported on the HAPPEI dataset are shown. The number in the first column indicates the rank of a performance, only those articles that are directly comparable are ranked together in a block. The first five articles that are fully comparable all report on the Root Mean Square

Error (RMSE) and both performance on the test set and validation set. Ranking is done based on the former, since the test set was not available at the time of submissions to the contest in which they all participated (EmotiW 2016) and will thus better indicate generalizability than the validation score. The next two studies only report on the performance on the validation set, and therefore cannot be ranked in the same block as the first five. Their performance is however still comparable to the validation performances of the first five. The two studies that follow perform cross-validation, and therefore get their performance from testing on different combinations of data. Four studies do not report on the RMSE but instead on the Mean Absolute Error (MAE), the last of which [38] not clarifying on which images testing is done. Sun *et al.* [64] report on the RMSE, MAE, and accuracy, and is therefore listed three times in this table. Huang *et al.* [35] report on the RMSE as well as the MAE.

6.2 GAFF Dataset

The GAFF dataset has two versions that are commonly used: GAFF 2.0 and GAFF 3.0. Two studies, however, work with a smaller version of these datasets. They are not directly comparable, as the images used may not be the same and they report on different metrics (Accuracy and Recognition Rate). All results are shown in Table 9, the specific GAFF version used in each study listed in the column named 'Version'. It should be noted that the GAFF 3.0 dataset is often referred to as GAFF 2.0, as it is an extension to that dataset. Some studies that claim to have used GAFF 2.0, while the number of images mentioned or the challenge they participated in indicates that they in fact used GAFF 3.0.

All studies that work with the GAFF 2.0 dataset have accuracy as the metric of choice. The first six studies report on both the test and validation performance, again being ranked on the former. An additional explanation may be in place for the performance of [83]: the scores reported (76.50 percent on the test set, 71.50 percent on the validation set)

TABLE 9
Results and Approach Details of Articles Working With the GAFF Dataset

#	Article	Metric	Perf.	Perf. on val	Version	Approach	Classic/CNN features
1	[9]	Acc	67.64%	-	Unspecified	Hybrid	Classic
1	[35]	RR	69.77%	-	Unspecified	Hybrid	Classic
1	[77]	Acc	80.90%	83.70%	2.0	Hybrid	CNN
2	[81]	Acc	80.60%	-*	2.0	Hybrid	CNN
3	[73]	Acc	79.78%	-*	2.0	Hybrid	Both
4	[83]	Acc	76.50%	71.50%	2.0	Hybrid	Both
5	[15]	Acc	66.34%	68.53%	2.0	Hybrid	CNN
6	[72]	Acc	64.68%	67.75%	2.0	Hybrid	CNN
1	[10]	Acc	-	82.30%	2.0	Hybrid	CNN
2	[63]	Acc	-	75.50%	2.0	Bottom-up	CNN
3	[71]	Acc	-	75.39%	2.0	Hybrid	CNN
4	[79]	Acc	-	74.00%	2.0	Hybrid	CNN
5	[1]	Acc	-	72.38%	2.0	Hybrid	CNN
6	[36]	Acc	-	72.17%	2.0	Bottom-up	Both
7	[75]	Acc	-	71.83%	2.0	Hybrid	CNN
8	[67]	Acc	-	55.23%	2.0	Bottom-up	CNN
1	[76]	Acc	77.76%**	-	2.0	Hybrid	CNN
1	[2]	Acc	68.08%	-*	3.0	Hybrid	CNN
2	[80]	Acc	67.48%	86.70%	3.0	Hybrid	CNN
3	[78]	Acc	65.59%	78.39%	3.0	Hybrid	CNN
4	[74]	Acc	64.83%	80.90%***	3.0	Hybrid	CNN
5	[82]	Acc	62.90%	76.30%	3.0	Hybrid	CNN
1	[42]	Acc	-	79.08%	3.0	Hybrid	CNN
2	[24]	Acc	-	65.70%	3.0	Hybrid	CNN

Shown are Performances on the Test Set and Validation Set.

** : Validation Performances Were Reported for Models Other Than the One Giving the Highest Test Result.*

*** : Test Set Randomly From Training and Validation Set.*

**** : Validation Performance on Part of the Validation Set.*

were achieved after the submission deadline of the contest they were participating in (EmotiW 2017). Their best performance that was submitted was 75.1 percent on the test set and 65.0 percent on the validation set. The goal of [63] was not to outperform the state-of-the-art, but to focus on a system that is applicable in real-time and still performs reasonably. The first six studies using GAFF 2.0 are followed by eight studies that only report on the validation score. Again, these cannot be ranked in the same block as the first six because the test score is not reported upon, but they can be compared based on their validation score. The last study in this table forms its own test set, randomly taken from a combination of the original training and validation sets.

The last GAFF version is the GAFF 3.0 dataset. Accuracy is the only metric used, and five out of the seven studies report on both the test and validation performance. It should be noted that [74] use only a part of the validation set for the validation performance. However, since ranking is done based on the test accuracy, it is still comparable to

the other studies inside that block. The last study using the GAFF 3.0 dataset, [24], shows an accuracy more than 10 percent lower than the other studies. However, their goal was not to outperform the state-of-the-art, but rather compare face-based emotion detection methods to non-face-based emotion detection methods. On GAFF 3.0 they only report on the non-face-based method, the comparison to face-based methods done with the EmotiC dataset, shown in Table 11.

6.3 MultiEmoVA Dataset

The last dataset that is employed more than once is MultiEmoVA, used in the two studies shown in Table 10. The performance noted for [34] is not the highest score to be found in their paper, as they also report on performances of arousal and valence detection separately. Here we show the highest performance of arousal and valence combined. As the two studies do not use the exact same images for testing ([36] performing cross-validation), they are not directly comparable.

6.4 Datasets Used Once

The last table, Table 11, lists all datasets that are used once by papers in this review. As none of them are comparable, the first column now simply indicates the row number. For the metric abbreviations used, we refer to Appendix A, available in the online supplemental material. The performance listed for the work of [30] is based on a positive/

TABLE 10
Results and Approach Details of Articles Working With the MultiEmoVA Dataset

#	Article	Metric	Perf.	Approach	Classic/CNN features
1	[34]	Acc	39.96%	Hybrid	Classic
1	[36]	Acc	54.40%	Bottom-up	Both

TABLE 11
Results, Approach, and Dataset Details of Articles Working With, for This Review, Unique Datasets

#	Article	Metric	Perf.	Dataset	Approach
1	[28]	Acc	98.54%	Crowd sound dataset	Top-down
2	[29]	Acc	99.24%	Crowd sound dataset (different from [28])	Top-down
3	[10]	Acc	89.1%	UCLA (protest)/UCLA (violence)	Hybrid
		Pc	0.91		
4	[30]	ME	0.412/0.305	Common-based/event-based	Bottom-up
5	[42]	Acc	89.93%	GroupEmoW	Hybrid
6	[20]	ARR	60.30%	People playing UNO	Bottom-up
7	[24]	Acc	88.40%	EmotiC	Hybrid
8	[4]	Z-score	-	NRC hashtag emotion corpus	Bottom-up
9	[21]	Acc	62.36%	People watching clips	Bottom-up
10	[22]	Acc	82.09%	People watching clips	Bottom-up
11	[25]	Acc	47.50%	VGAF	Top-down
12	[23]	Acc	0.812%/0.776%/ 0.831%/0.762%	FER2013/speech emotional corpus/ own twitter dataset/multi-modal	Bottom-up
		Prec	0.421/0.614/ 0.376/0.525		
13	[26]	Acc	90.9%/90.6%/100%	MED/ViolentFlows/UMN	Top-down
		Prec	97/91/100		
14	[16]	RR	>80%	People watching presentation	Top-down
15	[27]	AUC	0.993/0.9940	UMN/PETS2009	Top-down

neutral/negative distinction. In their paper they also report performance on a positive/negative distinction, leading to a higher performance. In [42], performance is reported on both the GAFF 3.0 dataset and the GroupEmoW dataset. For the former they work with face, scene, skeleton and object aspects, as was described in Section 5. The performance on the GroupEmoW dataset, however, results from a combination of face, scene and object aspects, leaving out the skeleton aspect. For [25], the performance listed is based on the combination of audio and video. In their paper, audio and video are also reported separately, audio reaching a better performance than audio and video combined. Nagarajan and Oruganti [24], also shown in the table of the GAFF 3.0 dataset, report on the accuracy of a place-based method, a scene-based method, and a face-based method. With this they aim to compare face-based methods to non-face-based-methods. The accuracy reported in the performance column, 88.40 percent, is their best performance. It is place-based, and the model is transfer learned on GAFF 3.0 data first. The other models reported on (using faces and scenes) are not transfer learned, so the models focusing on different aspects have to be compared to one another based on their performance without transfer learning. The accuracies are then 84.8, 85.1, and 85.5 percent for the places, scenes, and faces models, respectively. This shows that their place-based methods performs worse than the other two, but only by a very small margin.

7 DISCUSSION

This section provides a discussion on topics described in this review. First the performances that were listed in the previous section are discussed, on the basis of approaches taken, features used, and comparability of the reported results. This will answer the fourth subquestion of this review: *are performances comparable across studies?* Thereafter the application possibilities of the frameworks described in

this review are discussed, examining characteristics of real-world groups and to what extent these are incorporated in current literature. This will focus on answering the fifth and final subquestion of this review: *to what extent are the discussed approaches applicable to real-world groups?*

7.1 Performances

7.1.1 Approaches Taken

Inspecting all tables, an interesting finding is that the majority of best performances (where comparison is possible) take a hybrid approach. Looking at Table 8, [70] perform best of the studies reporting both test and validation results. Ghosh *et al.* [10] outperforms their validation accuracy, but does not report on the test accuracy. Both take hybrid approaches, as do [35] (RMSE, cross-validation, and MAE, cross-validation). For the GAFF 2.0 dataset, the best performance comes from [77]. Of the studies reporting on the validation accuracy only, [10] performs best with an accuracy 1.6 percent below that of [77]. Both have a hybrid approach. Lastly, for the GAFF 3.0 dataset, the best performance comes from [2]. With their hybrid approach they won the GREco EmotiW 2018 subchallenge. The second best result is from the hybrid approach of [80], their validation performance outperforming that of [2]. Fig. 10, like Fig. 2, shows the number of articles published per year, but here each year is divided into bar colours based on the approach taken. It can be seen that hybrid approaches start appearing in 2015 and become increasingly popular in the following years. Up until 2019, they are used more often than both bottom-up and top-down approaches separately, but in 2019 they are used as often as bottom-up approaches. This may be explained by the absence of the GREco EmotiW subchallenge in 2019, after being organised in the three preceding years (also explaining the drop in the absolute number of articles published). This leads to a reduced number of studies working with image data, relatively increasing the

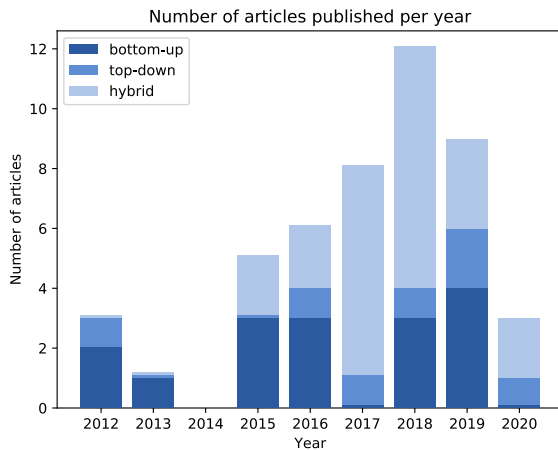


Fig. 10. Shown is the number of articles published per year. The colours within each bar indicate the number of articles that use a particular approach (bottom-up, top-down, or hybrid) each year.

number of studies working with other modalities: when working with social media text messages, for example, any approach will be bottom-up by definition [23], [30]. The tendency of authors in recent years to use hybrid approaches, when possible, may be explained by the good performances resulting from them.

7.1.2 Classic/CNN Features

The tables for image datasets also include a column ‘Classic/CNN features,’ mentioning for each study if features are extracted with classic descriptors, with CNNs, or with a combination of both. From Table 8 and the studies using the GAFF 2.0 version in Table 9, there seems to be no clear answer as to which of the three gives the best performance. For different blocks, either the combination of both outperforms ‘Classic’ and ‘CNN,’ ‘CNN’ outperforms the combination of both, or ‘Classic’ outperforms the combination of both. The other image datasets are used with only classic features (the unspecified GAFF versions in Table 9), with only CNN features (studies using the GAFF 3.0 version in Table 9), or with only one type per comparable study (Table 10). As mentioned in Section 5, it becomes clear that neural approaches are becoming increasingly popular in recent years. Despite their popularity and benefits such as computational power and ability to generalize, (convolutional) neural networks are not necessarily the single best approach towards group emotion detection: as becomes clear from the previously discussed tables, in some cases a framework benefits from a combination of neural networks and classic descriptor features, rather than just the former. This potentially mutually beneficial relationship between classic approaches and CNN approaches should be further explored.

7.1.3 Comparability

Many of the studies described in this review are not directly comparable to other studies: either because they report on different metrics, use different parts of a dataset, or because they use entirely different datasets. For some research, a different or new dataset is necessary to fit the goal of that particular study. However, for the sake of clear, replicable, and comparable studies, we believe it is necessary for researchers

to work with existing and commonly used datasets and metrics as much as possible, and to clearly state which parts of the dataset are used for training, validating and testing.

7.2 Application Possibilities

This review has described the literature on automatic group emotion recognition in terms of group types and emotion models, datasets, and approaches. The majority of articles discussed here validate their frameworks with existing data, however, this does not automatically guarantee the frameworks to be generalizable to uncontrolled, real-world environments. This section will discuss the application possibilities of the presented frameworks. To this end, we assume a real-world group of people to have the following characteristics:

- 1) A group can change in its composition, with people joining or leaving what is considered the group.
- 2) A group can be heterogeneous with respect to its emotions, containing different emotional subgroups.
- 3) The emotion(s) of a group can change over time.
- 4) Behaviour of group members can change due to reasons other than emotional ones. Sounds emerging from a group, for example, can appear or disappear without an emotional cause for that.

The first characteristic implies that a particular group, in need of analysis, can become smaller or bigger during the time in which that group is to be monitored. Imagine for example a conference: shortly after opening there may only be a small number of people, while during an important talk there may be hundreds of them in the same room. However, methods used to recognize emotions from small groups differ from those to analyse large groups. Techniques focusing on individuals, through facial or conversation analysis, may not work on a larger scale when individuals (either their faces or their voices) get lost in the mass. An additional component to consider is the computational intensity of such an individual-based method when applied to large groups. Reversely, when considering large gatherings, the opposite of the pitfall of small groups may occur. A model trained to recognize the movements of large crowds, or the sounds that emerge from masses, is likely to fail to recognize the movements and sounds of a small gathering. Textual data from social media has the potential to do well in both small and large groups, as each individual message can be analysed irrespective of the number of messages or the physical dynamics within the group.

The majority of the literature discussed in this review considers a group under analysis to have one common emotion or a common combination of emotions. This means that their frameworks are built to find that common emotion or combination of emotions. For the challenges faced by the discussed research this suffices, however, it reduces the application possibilities, as real-world groups are not likely to consist of individuals sharing the same emotions in the same period of time. Many studies classify images showing a small number of people. This could potentially be used to infer an emotion distribution of a larger group, by analysing multiple images of multiple subgroups simultaneously. Analysis of the data modalities video and social media may

also be flexible in this regard, provided that the social media messages contain a geo-location. Other data modalities, such as audio, are less likely to analyse parts of a group in their current form. However, more research is required to explore possibilities.

The third real-world group characteristic states the changing of emotions over time. The majority of studies in this review classify emotions at one point in time. Some exceptions to this are the works of [20] (face and speech analysis), [4] (social media analysis), and [27] (video analysis). While studies that classify emotions only at a single point in time could potentially be used for emotion tracking as well, by performing analysis at multiple consecutive moments, this is in need of more research.

The final group characteristic given here states that group behaviour can change due to non-emotional reasons. This is of importance as most current methods rely solely on one data modality and the variations that can be distinguished within that data modality. When working with video data to analyse emotions through the motions of the group, for example, one can think of a scenario where a group changes its motion because of orders given by security guards. Similar scenarios can be thought of with other data modalities, making a prediction relying on that data less reliable. Hybrid approaches, considering both bottom-up and top-down components of a data modality, could improve this reliability by basing their prediction on multiple components. In this review, hybrid approaches were only taken with images. More work needs to be done to investigate the performance of hybrid approaches working with data modalities other than images, so as to keep the benefits of such an approach when working in a scenario where image collection is not feasible. Additionally, multi-modal approaches have the potential to improve the reliability of prediction methods, by means of complementarity [86]. In the example of changing movement due to security guard orders, it is not likely that the sudden change is reflected in all other data modalities as well. The only multi-modal works in this review are those of [20] and [23], indicating a need of more research in this area.

Having discussed the characteristics of real-world groups, the question remains if these characteristics are represented in existing datasets. A framework may be theoretically able to cope with all described characteristics, while still not performing as required in a real-world context due to the data the model was trained on. Ideally, data should be annotated without the annotation process influencing the (labelled) data, and collected without the collecting itself influencing the data. The former issue was raised in a recent review by Dudzik *et al.* [87], describing how an annotator (perceiver of the emotion data) can be biased in his or her interpretation of other people's emotions. Regarding the latter issue, an example of the ideal situation could be to search for social media messages related to an event, where the collection itself will not influence the content of these messages. Audio and video recordings are collected onsite. As they will be recorded continuously over a period of time, the influence the recording itself will have on the data is not likely to be significant. Image datasets are of a different nature, as the people in images will typically be aware that an image is taken, or they might even pose for it. This

reduces the real-world application possibilities of such datasets and introduces possible biases. At the same time, images are a rich source of information for emotion estimation, explaining the popularity of image-based emotion recognition. To the best of our knowledge, this issue has not yet been tackled in the literature, leaving room for the development of new and improved datasets.

8 CONCLUSION

In this review, the aim was to describe what researchers have done in recent years to investigate automatic group emotion recognition. To the best of our knowledge, we are the first to provide a comprehensive overview on group emotion estimation that covers a wide range of subjects, from group types and emotion models to performances. This work can therefore be an interesting read to those working in the field, or a good base for those interested in starting.

The main question of this review was *what is the current state of the field of automatic group emotion recognition?* This question was answered through five subquestions, which were discussed in Sections 3 through 7. Before embarking on these subquestions, articles were selected with Dense Query Extraction and automatic cross-referencing (Section 2), including 47 studies. A summary will be provided on each subquestion, outlining where and how it was answered.

- 1) *What types of groups (dynamics, size, emotions) are considered?*

Section 3 focused on the first subquestion by describing the types of group and emotion models considered in the literature. Group types ranged from small, static group to dynamic mass gatherings. Emotions are categorised differently in the research field, but categorisations can be roughly divided in derivatives of discrete emotion labels [31], derivatives of arousal-valence emotion planes [32], and studies focusing on one type of emotion.

- 2) *What datasets are common in the field of group emotion research?*

This subquestion was answered in Section 4, dividing the datasets used in the literature into four categories: group emotion image datasets, facial emotion and facial recognition datasets, video datasets, and datasets of other data modalities. For each category, all common datasets are discussed.

- 3) *What bottom-up and top-down approaches exist for different data modalities?*

Methods used by authors to classify emotions, either from a bottom-up perspective, a top-down perspective, or a hybrid combination of the two, were discussed in Section 5. This section elaborated on the types of features used for analysis per approach and per data modality, roughly distinguishing between classic descriptors, neural descriptors and combinations thereof. Additionally, Section 5 discussed the methods for fusion of different modalities, fusion of different analysis aspects (for example combining facial and scene analysis) and fusion of individual

features and emotions to determine the common group emotion.

4) *Are performances comparable across studies?*

The foundation for answering this subquestion was laid in Section 6, listing all performances, grouped per dataset for comparability. Discussion of these results was done in Section 7. For the modality of images, hybrid approaches showed better performances than both bottom-up and top-down approaches separately. With other modalities, such an approach has not yet been taken. Classic, non-neural approaches should not be fully disregarded, despite the popularity of neural approaches: the results discussed in Section 6 show that a high performance in some cases comes from combining classic with neural approaches, instead of only using the latter.

Many of the studies in this review are not directly comparable to one another. This has multiple causes: they report on different metrics, use different parts of a dataset for testing, or a different dataset in general. For the sake of clear, replicable, and comparable studies, we believe it is necessary for researchers to work with existing and commonly used datasets and metrics as much as possible, and to clearly state which parts of the dataset are used for training, validating and testing. We therefore want to suggest to report on multiple common metrics, such as the accuracy and RMSE, as well as on multiple common datasets suited for the task. The latter also serves the purpose of showing the generalizability of a proposed model: the more homogeneous the results are across datasets, the less overfitting will have taken place during training. Some studies provide a highly detailed description of the implementation of their system, however, this does not yet seem to be common practice: we strongly urge researchers to share all details needed to replicate a study, and to make code publicly available when possible.

5) *To what extent are the discussed approaches applicable to real-world groups?*

Section 7 elaborated on application possibilities, providing an interesting insight into which methods outperform others and to what extent current work can be applied in the real world. While techniques for automatic group emotion recognition keep improving and show impressive results, the real-world applicability of such methods is not always guaranteed. Characteristics of real-world groups are not commonly incorporated in current frameworks. Suggestions for future work in this direction will be given next.

Future Work. For future work, we suggest working towards a common practice of solid reporting on methodology and results, as it has become clear that this is not always the case. Next to methodological improvements, we think that important work lies in improving the real-world applicability of current methods. We therefore propose to work towards systems that are able to cope with real-world group characteristics (as described in Section 7.2).

The first of these is changing group size. A challenge for future work lies in building methods that are flexible in

the group size to analyse. An example of this could be a network that automatically detects the group size and changes the analysis accordingly.

The second real-world characteristic not yet incorporated in current research, is that of having different emotional subgroups. When investigating a real-world group, the group may be divided into smaller subgroups that share a common emotion, possibly different from that of other subgroups. Future work could therefore focus on finding and analysing different emotional subgroups. A concrete example could be to detect subgroups whose emotions highly deviate from the average emotion, such as people getting angry during an event.

Third, as discussed in Section 7.2, only three studies in this review analyse changing emotions over time. Temporal analysis and the challenges this brings could therefore be addressed in future work. Temporal analysis can be useful for detecting patterns that cannot be detected with data from a single point in time, which can in turn help to predict emotions. This could for example be adopted for simulations of events or other gatherings, in which security guards are trained to handle different (emotional) situations, with direct feedback on their actions.

The fourth and final real-world group characteristic not yet sufficiently addressed in research, is robustness to variations in data not caused by an emotional change. These possible biases should be further investigated.

Furthermore, a concrete starting point for future work could be to explore the feasibility of hybrid approaches that work with data modalities other than images and further exploring multi-modal approaches, for example combining modalities such as images and audio to complement each other. Additionally, to allow studies to work towards more real-world applicable frameworks, datasets should be created that aim to reduce any biases that may arise from the data collection itself.

ACKNOWLEDGMENTS

This work was part of the research programme Innovative Research Incentives Scheme Vidi SSH 2017 with project number 016.Vidi.185.178, which is financed by the Dutch Research Council (NWO).



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Emmeke A. Veltmeijer received the BSc degree in psychobiology, and the MSc degree in artificial intelligence from the University of Amsterdam, in 2019. This sparked an interests include the interface between humans and technology. She is currently working toward the PhD degree with the Social AI Group, Vrije Universiteit Amsterdam. Her research interest her research she focuses on affective computing in large groups.



Charlotte Gerritsen is currently an assistant professor within the Social AI group, Vrije Universiteit Amsterdam. She is the recipient of a Vidi grant, a prestigious personal grant in the Netherlands. Her research is located at the intersection of artificial intelligence and the social sciences. She aims to enrich computational agent based models with knowledge of human behaviour in order to make these models more realistic. These models are used to understand, predict, and influence human behaviour in various domains.



Koen Hindriks is currently the head of the Social AI group, Vrije Universiteit Amsterdam and a co-founder of the startup Interactive Robotics. He also co-chairs the Special Interest Group on Artificial Intelligence in The Netherlands. His work on Artificial Intelligence focuses on smart social interfaces, cognitive agent technology, and socio-cognitive robotics. He has developed an expressive emotion engine for robots and designed robots as companions and assistants for health care and education.

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